

Section A: Structured Questions

Answer **all** the questions in the spaces provided.

- 1 (a) Fig. 1.1 shows the second ionisation energies of eight elements with consecutive proton numbers and one of the elements is aluminium.

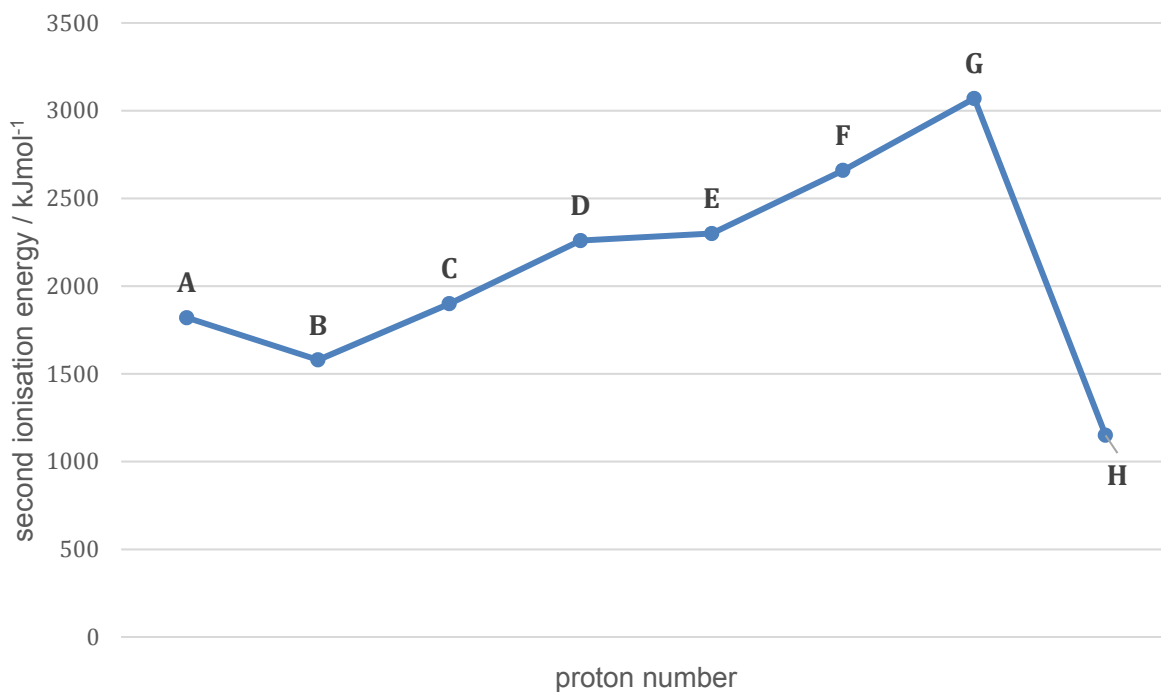


Fig.1.1

- (i) Define, with the aid of an equation, what is meant by the second ionisation energy of aluminium.

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 [2]

- (ii) Explain why the second ionisation energy of aluminium is usually more endothermic than the first ionisation energy.

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 [2]

(iii) Identify the element which represents aluminium. Explain the reasoning.

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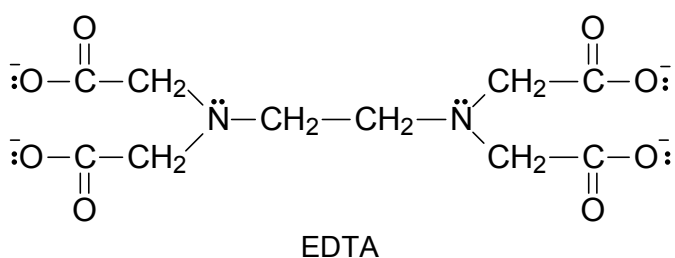
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(b) Hard water is high in dissolved minerals, specifically calcium ions. Although hard water does not pose a health risk, it causes poor soap performance. The soap used in hard water combines with the dissolved minerals to form a sticky soap curd which consists of insoluble salts.

(i) Ethylenediaminetetraacetate or EDTA is a component in laundry detergent.



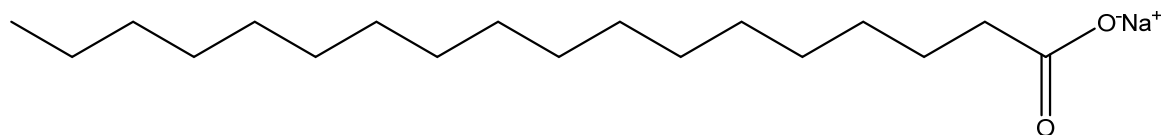
Suggest how EDTA could improve the soap performance in hard water.

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- (ii) Soap has effective cleaning properties because it contains one or more surfactants.

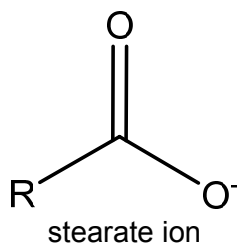
A surfactant is a molecule with a *non-polar* tail and a *polar* head. An example is sodium stearate, as shown below.



A simplified representation of stearate ion is RCOO^- .

Complete the diagram by drawing the relevant dipoles and indicate the type of intermolecular forces of attraction between stearate ion, an oil molecule and a water molecule. Hence explain how stearate ion helps to remove oil from surface during washing.

You may use R' to represent an oil molecule.



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[3]

[Total: 10]

- 2 The table below shows the melting points of some of the chlorides of the elements in period 3 of the periodic table.

Chlorides	Melting points/ °C
Magnesium chloride, MgCl_2	714
Aluminium chloride, AlCl_3	192
Phosphorus pentachloride, PCl_5	161
Sulfur dichloride, SCl_2	-121

- (a) Account, in terms of structure and bonding, the difference in the melting points between

- (i) magnesium chloride and aluminium chloride

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- (ii) phosphorus pentachloride and sulfur dichloride

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[2]

- (b) When PCl_5 is added to water, liquid phosphorus oxychloride, POCl_3 is formed as an intermediate compound.

- (i) Reactions of covalent chlorides with water can be rationalised as step-wise replacement of $-\text{Cl}$ with $-\text{OH}$. Deduce a three-step reaction sequence for the formation of POCl_3 from PCl_5 .

Step 1:

Step 2:

Step 3:

[3]

POCl_3 can also act as a *Lewis base* to remove the AlCl_3 at the end of a Friedel–Crafts reaction, resulting in the formation of a covalent product.

(ii) Define the term '*Lewis base*'.

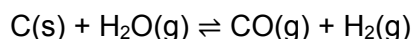
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[1]

(iii) Draw a diagram to illustrate the shape of the product which results in the removal of AlCl_3 , stating the type of bonding present and the bond angles around the central atoms.

Type of bonding:
[2]

[Total: 10]

- 3 (a) When steam is passed over hot coke (a form of carbon obtained from coal), water gas is produced. Water gas, consists of a mixture of carbon monoxide and hydrogen, is an important industrial fuel.



When 0.100 mol of steam was reacted with coke in a vessel of 1.00 dm^3 and allowed to reach equilibrium at 800°C , the partial pressure of steam was found to be 2.67 atm.

(i) Calculate the initial pressure of steam, in atm, introduced into the vessel.

[2]

- (ii) Using your answer in **a(i)**, calculate a value for K_p , in atm, at 800 °C.

[2]

- (iii) Calculate the minimum mass of carbon required in the vessel.

[2]

- (iv) State and explain whether high or low pressure would favour a higher yield of water gas.

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[2]

- (v) The reaction was repeated at a lower temperature and the numerical value of K_p was found to have decreased.

State and explain whether the forward reaction is endothermic or exothermic.

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[2]

- (b) (i) The relationship between ΔG° and K_p for the reaction between carbon and steam is

$$\Delta G^\circ = -RT \ln K_p$$

where ΔG° is the standard Gibbs free energy change in J mol^{-1} , R is the gas constant, T is the temperature in Kelvin at which the equilibrium is established and K_p is the equilibrium constant.

Given that the K_p of the reaction at 298 K is 2.97×10^{-18} atm, calculate the ΔG° of the reaction.

[1]

- (ii) Comment on the sign of ΔG° with reference to the position of equilibrium of the reaction.

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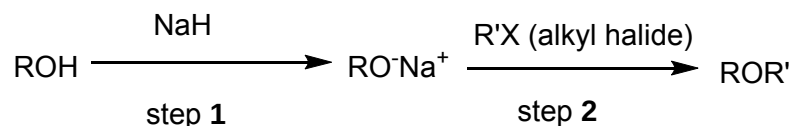
[2]

[Total: 13]

4. (a) Ethers have the general formula, R-O-R' (where R and R' are alkyl or aryl groups).

The most useful method of preparing ethers is by the Williamson ether synthesis, in which an alkoxide ion (RO^-) reacts with an alkyl halide ($\text{R}'\text{X}$) in an $\text{S}_{\text{N}}2$ mechanism.

A reaction scheme is shown below:



- (i) Suggest why step 1 is necessary in the Williamson ether synthesis.

[1]

- (ii) Using 1-chloropropane as the alkyl halide and $\text{CH}_3\text{CH}_2\text{O}^-$ as the alkoxide ion, outline the mechanism for step 2 of the Williamson ether synthesis. Show all charges and relevant lone pairs of electrons and show the movement of electron pairs by using curly arrows.

[3]

- (iii) The same reaction in a(ii) was repeated using iodopropane instead of chloropropane. Using relevant data from the *Data Booklet*, state and explain the effect on the rate of reaction.

[2]

- (iv) Suggest why the product obtained in **a(ii)** shows no optical activity.

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[1]

Fig. 4.1 shows two separate reactions to synthesise ethers using the Williamson ether synthesis method.

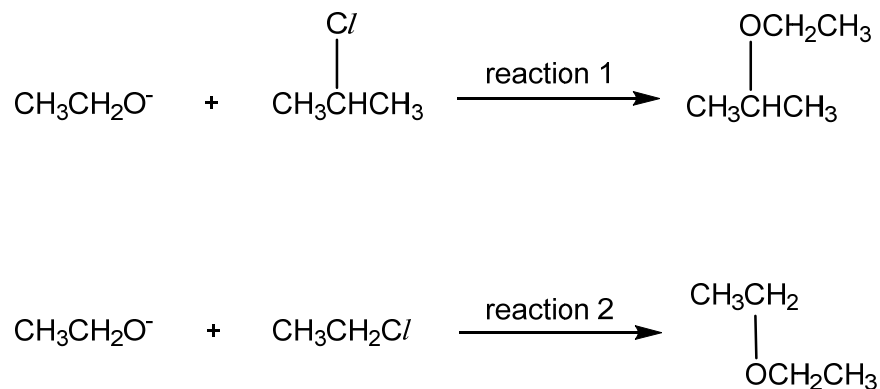


Fig 4.1

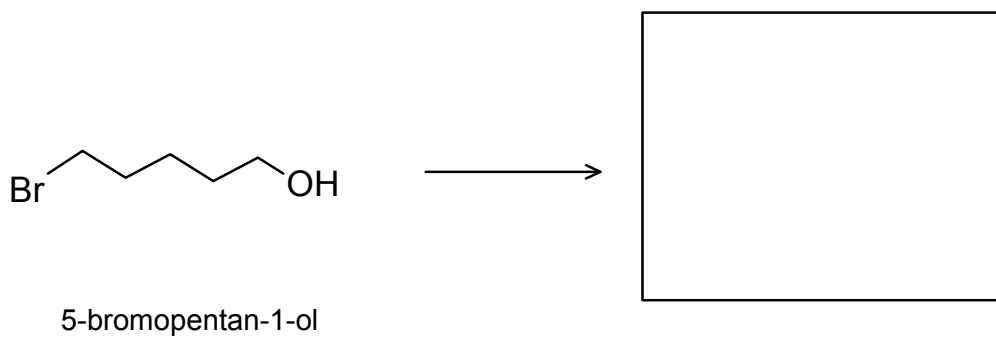
- (v) State and explain which reaction will give a higher yield.

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[2]

- (vi) Suggest the structure of the product when 5-bromopentan-1-ol undergoes the Williamson ether reaction.

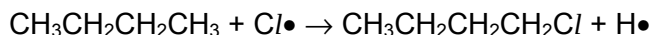


[1]

- (b) Halogenated organic compounds can be synthesised from various functional groups.

Chlorofluorocarbons (CFCs) have been widely used in aerosol sprays, refrigerants, but are now known to destroy ozone in the upper atmosphere through a free radical chain reaction.

- (i) A student wrongly proposed that a propagation step for the formation of 1-chlorobutane from butane to be:



Write the correct propagation step. Hence use relevant data from the *Data Booklet*, calculate the enthalpy change of reaction to explain why the propagation step proposed by the student is incorrect.

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[3]

- (ii) 2,5-dimethylhexane is reacted with chlorine in the presence of UV light to form 3 monochlorinated alkanes.

It has been found experimentally that in free radical substitution of alkanes, the primary, secondary and tertiary hydrogen atoms are replaced by chlorine atoms at different rates, as shown in the following table.

type of hydrogen atom	reaction	relative rate
primary	$\text{RCH}_3 \rightarrow \text{RCH}_2\text{Cl}$	1
secondary	$\text{R}_2\text{CH}_2 \rightarrow \text{R}_2\text{CHCl}$	7
tertiary	$\text{R}_3\text{CH} \rightarrow \text{R}_3\text{CCl}$	21

Draw the structural formula of the 3 possible monochlorinated products of 2,5-dimethylhexane and using the information in the table, suggest the relative ratios in which they are formed.

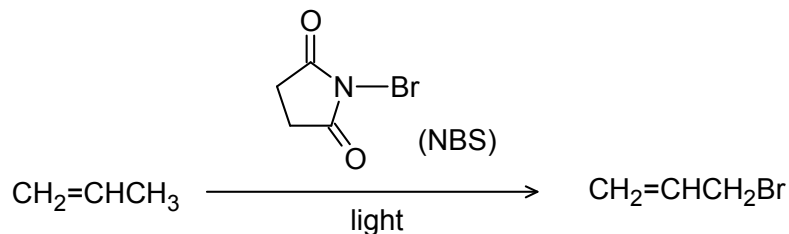
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Ratio: : :

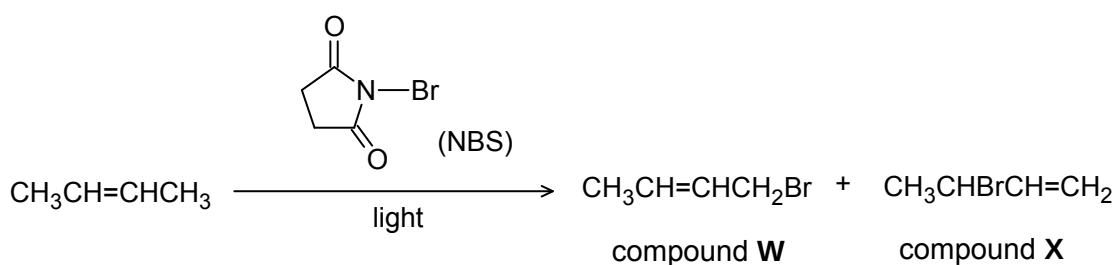
[3]

- (c) Another method of preparing alkyl halides from alkenes is by the reaction with N-bromosuccinimide (NBS) in the presence of light to give products which are substituted at the allylic position i.e. the position *next* to the double bond. NBS is used as a source of bromine.

The process is similar to the free radical substitution mechanism of alkanes.



- (i) However, a mixture of 2 products (**W** and **X**) is formed when but-2-ene undergoes this reaction.



Suggest a reason for the formation of compound **X**.

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[2]

- (ii) State the type of isomerism shown by compound **W** and compound **X**.

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[1]

[Total: 19]

- 5 Infra-red(IR) absorptions can be used to identify functional groups in organic compounds. For example, ethyl ethanoate shows absorption at $100 - 1300 \text{ cm}^{-1}$ and $1680 - 1750 \text{ cm}^{-1}$.

A student used infra-red spectrometer to obtain an infra-red(IR) spectrum of benzyl alcohol $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ as shown in Fig. 5.1.

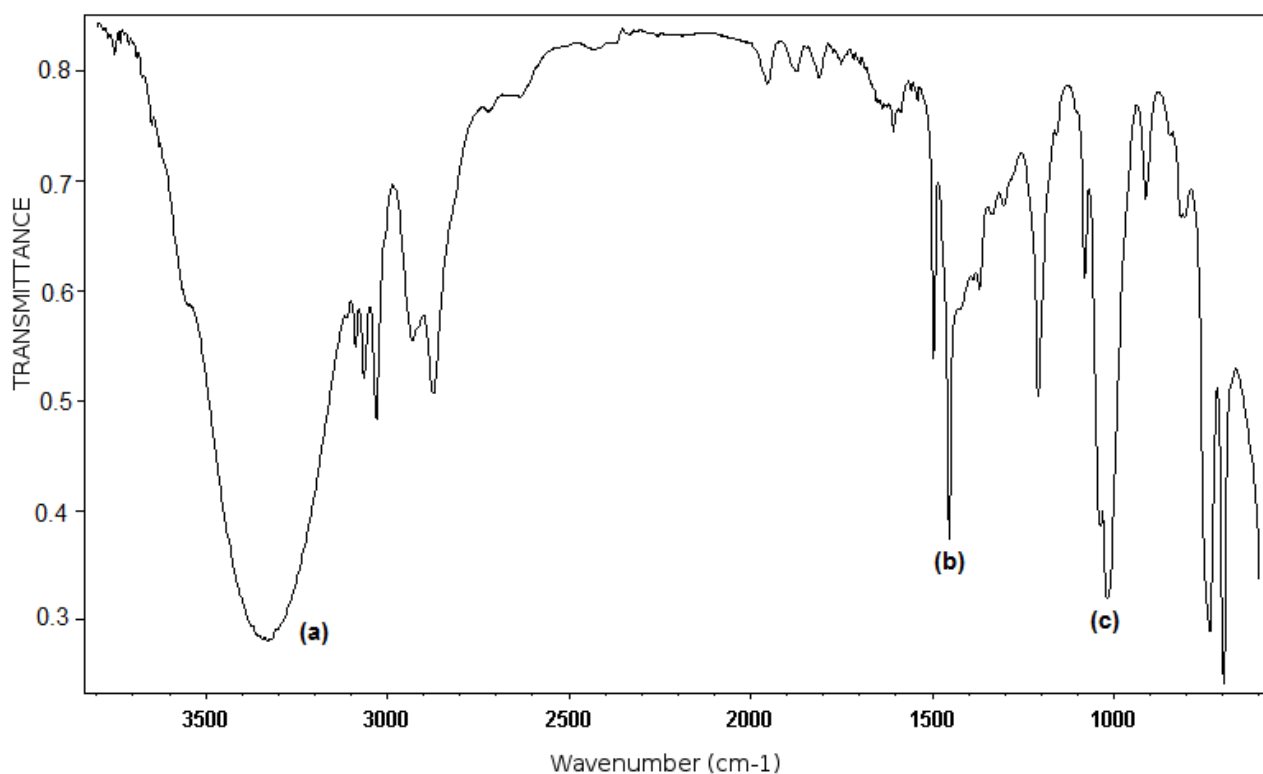


Fig. 5.1

- (a) Use the table of characteristic infra-red absorption frequencies in the *Data Booklet* to identify the infra-red absorption range shown by benzyl alcohol.

IR region	Wavenumbers/ cm^{-1}	Functional groups present
(a)		
(b)		
(c)		

[3]

- (b) The student carried out reflux of benzyl alcohol with acidified potassium dichromate and the product **A** obtained was analysed using IR spectrometer. Fig. 5.2 shows the IR spectrum obtained.

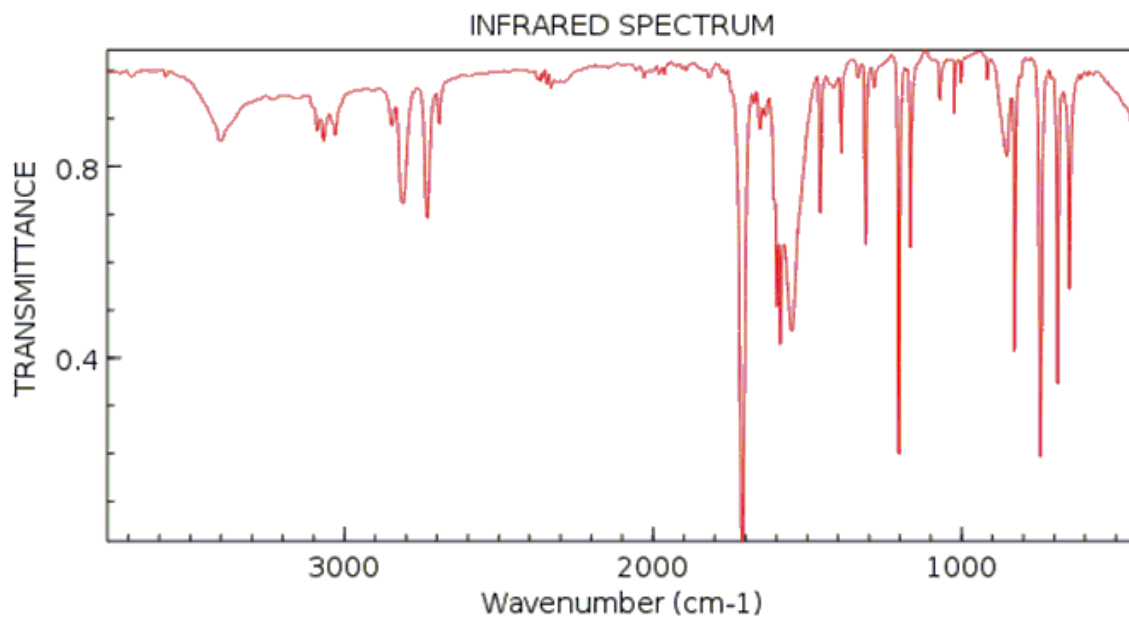


Fig. 5.2

By comparing the two infra-red spectra, suggest two differences from the infra-red spectra and draw the structure of the product **A** obtained.

Product **A**:

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[3]

- (c) Suggest a simple chemical test that could be used to confirm benzyl alcohol $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ has been completely converted into product **A** in (b).

Test :

Observations:

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[2]

- (d) A student wishes to determine the concentration of product **A** obtained. The student prepares 5 solutions of known concentrations, 0.0100, 0.0200, 0.0400, 0.0800 and 0.100 mol dm⁻³ of product **A**. 10 μ L of each solution with the sample was used and a high performance liquid chromatogram was obtained. The peak areas of the chromatographs are given in the table below and a graph of concentration against peak area was obtained in Fig.5.3.

Solution concentration / mol dm ⁻³	0.0100	0.0200	0.0400	0.0800	0.100	sample
Peak area / cm ²	0.50	0.83	1.39	2.63	3.18	2.53

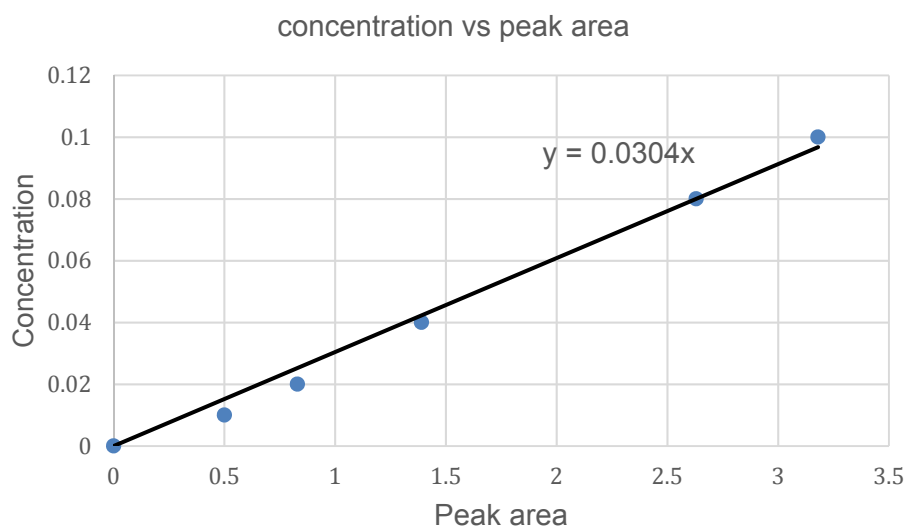


Fig. 5.3

- (i) Using the peak area of the sample from the table and information from the graph, determine the concentration of product **A** in the sample.

[1]

- (ii) Suggest what the student can do to ensure high yield of product **A** from the reaction of benzyl alcohol with acidified potassium dichromate in (b). Use Le Chatelier's Principle to explain your suggestion.

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[2]

(e) Fig. 5.4 shows a Ritter reaction.

The Ritter reaction transforms a nitrile to amide using a strong acid and water as follows.

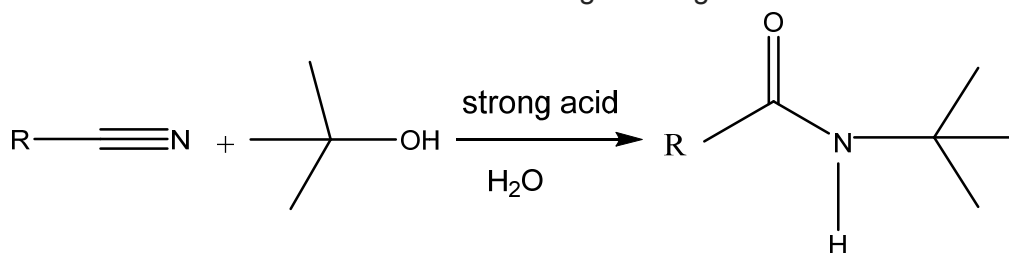
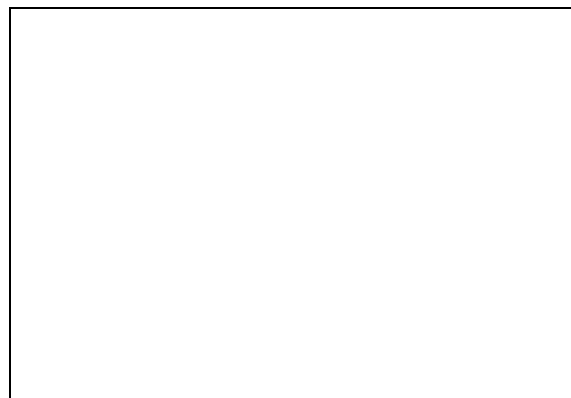
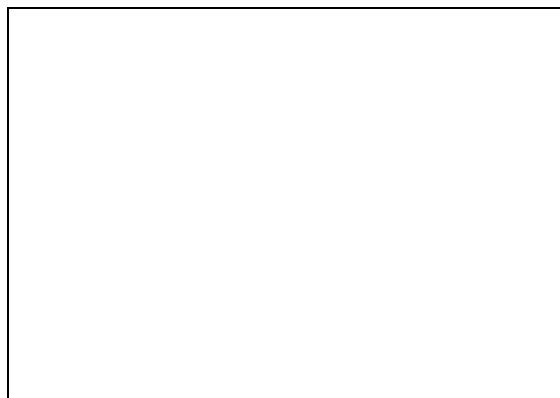


Fig. 5.4

(i) Side products will be formed from the Ritter reaction if heating was carried out.

Suggest the type of reaction which occurs and draw the structures of two side products formed.

Type of reaction:



[3]

(ii) Benzyl alcohol, $\text{C}_6\text{H}_5\text{CH}_2\text{OH}$ undergoes Ritter reaction with CH_2CHCN as shown in Fig. 5.4. Draw the structure of the amide formed.

[1]

- (f) Fig. 5.5 shows a possible process in the human body where benzyl alcohol is oxidised to benzoic acid, conjugated with amino acid, glycine in the liver, and excreted as benzoylaminoethanoic acid, as shown.

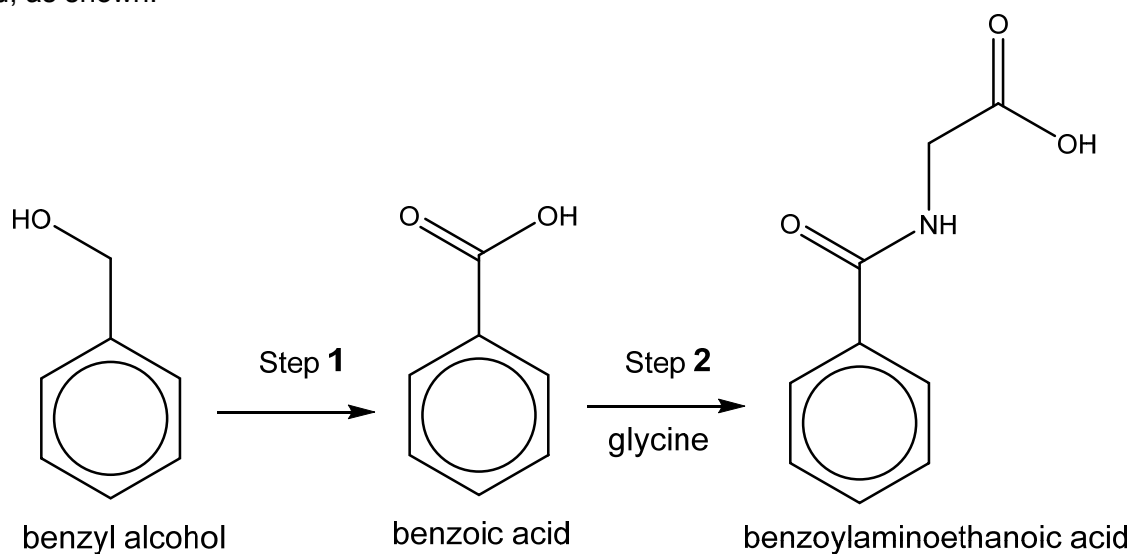


Fig 5.5

- (i) Write an equation for the oxidation of benzyl alcohol to form benzoic acid. Use [O] to represent the formula of the oxidising agent.

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[1]

- (ii) What type of reaction takes place in step 2?

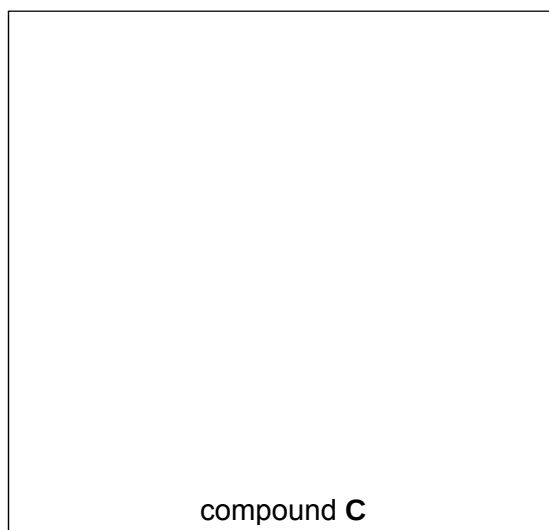
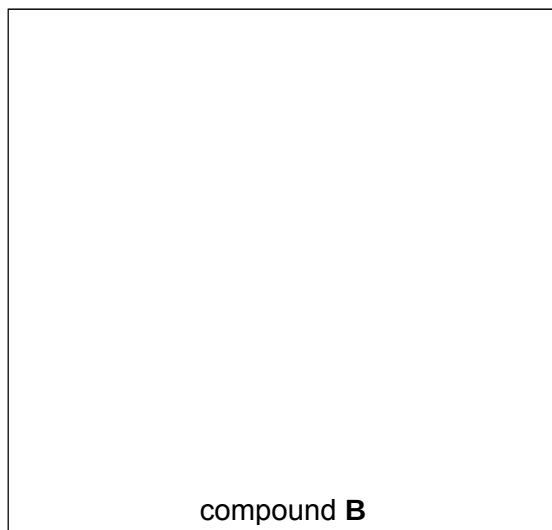
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[1]

- (iii) Draw the displayed formula of a tripeptide consisting of glycine only.

[1]

- (iv) Benzoylaminoethanoic acid can be reduced by LiAlH_4 in dry ether at room temperature and hydrogen gas with nickel catalyst under high heat and pressure. The organic products formed in the two reactions are different. Compound **B** has a molecular mass of 151 and compound **C** has a molecular mass of 185.

Deduce the structures of **B** and **C**.



[2]

The synthesis of glycine in the liver is catalysed specifically by glycine synthase enzyme.

- (v) Describe, with the aid of a sketch how the rate of synthesis of glycine is affected by the concentration of the substrate.



[3]

[Total: 23]